

Financial Risk Management

Mapping oil price risk, for a portfolio management strategy

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Understanding oil price shocks

American Economic Review 2009, 99:3, 1053–1069
<http://www.aeaweb.org/articles.php?doi=10.1257/aer.99.3.1053>

Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market

By LUTZ KILIAN*

.....Motivation for Indian Economy?

Understanding oil price shocks

Oil supply shock

Aggregate demand shock

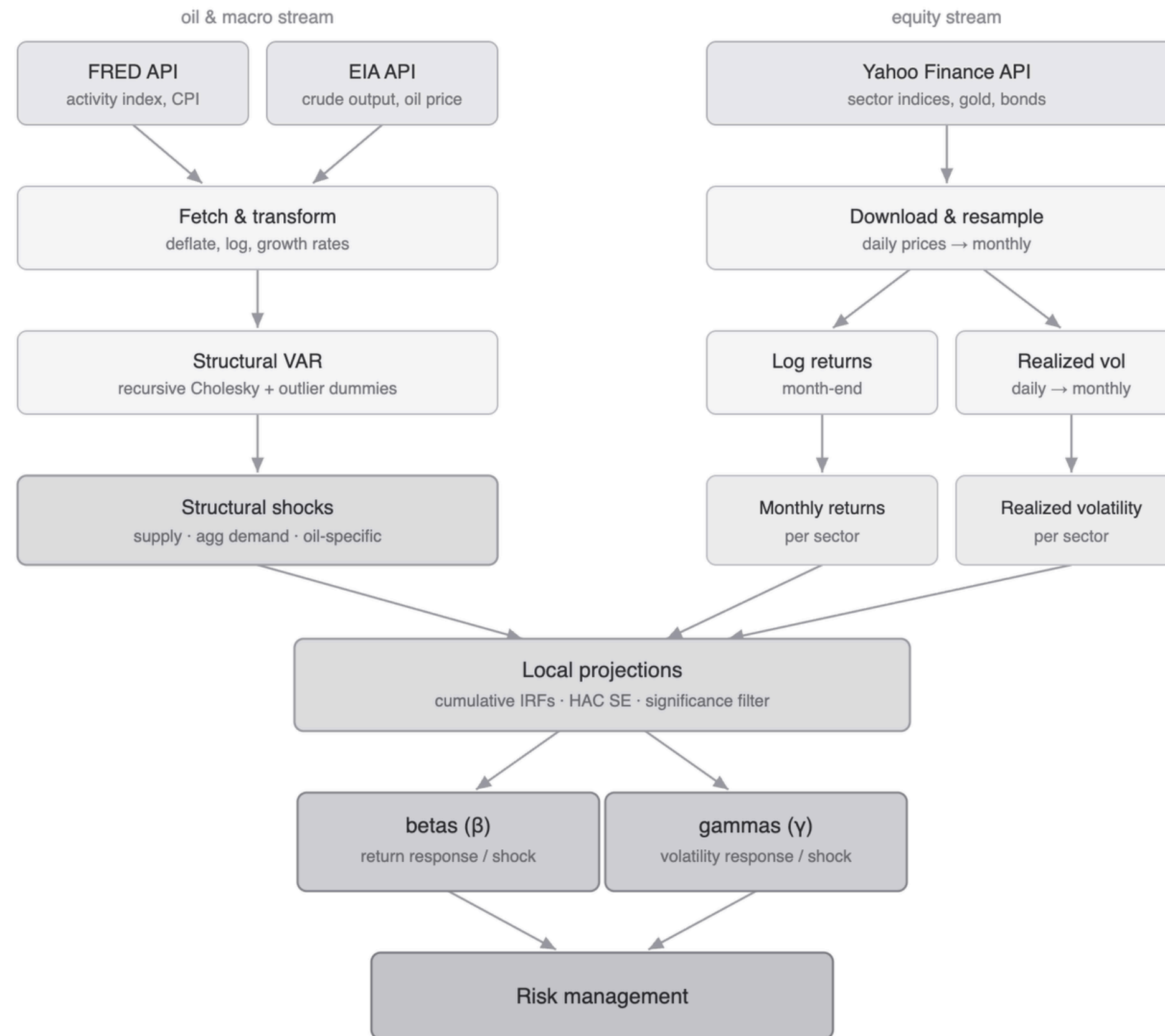
Oil-specific demand shock

Structural VAR

$$\begin{bmatrix} \Delta prod_t \\ rea_t \\ rpo_t \end{bmatrix} = c + A_1 \begin{bmatrix} \Delta prod_{t-1} \\ rea_{t-1} \\ rpo_{t-1} \end{bmatrix} + \cdots + A_p \begin{bmatrix} \Delta prod_{t-p} \\ rea_{t-p} \\ rpo_{t-p} \end{bmatrix} + u_t$$

$$\mathbf{e}_t \equiv \begin{pmatrix} e_t^{\Delta prod} \\ e_t^{rea} \\ e_t^{rpo} \end{pmatrix} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{pmatrix} \varepsilon_t^{oil\ supply\ shock} \\ \varepsilon_t^{aggregate\ demand\ shock} \\ \varepsilon_t^{oil\ specific-demand\ shock} \end{pmatrix}$$

Data Pipeline



Computing the Shocks

1. Reduced-form VAR

Each variable on its own & others' lags

2. Residuals are correlated

Off-diagonals $\neq 0 \rightarrow$ causes entangled

3. Cholesky decomposition

Lower-triangular P encodes recursive ordering

4. Orthogonalized shocks

Unit-variance, mutually uncorrelated

$$z_t = \alpha + A_1 z_{t-1} + \dots + A_p z_{t-p} + u_t \Rightarrow \Sigma = \text{Cov}(u_t) = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_2^2 & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 \end{pmatrix} \Rightarrow \Sigma = PP', \quad P = \begin{pmatrix} p_{11} & 0 & 0 \\ p_{21} & p_{22} & 0 \\ p_{31} & p_{32} & p_{33} \end{pmatrix} \Rightarrow \varepsilon_t = P^{-1}u_t, \quad \text{Cov}(\varepsilon_t) = I_3$$

The recursive structure, written out: $u_t = P\varepsilon_t$ — each shock “turns on” one new variable

$$u_{1t} = p_{11} \varepsilon_t^{\text{supply}}$$

$$u_{2t} = p_{21} \varepsilon_t^{\text{supply}} + p_{22} \varepsilon_t^{\text{agg.dem}}$$

$$u_{3t} = p_{31} \varepsilon_t^{\text{supply}} + p_{32} \varepsilon_t^{\text{agg.dem}} + p_{33} \varepsilon_t^{\text{oil.spec}}$$

Oil supply shock

Unexpected change in oil production (e.g. a disruption)

Aggregate demand shock

Surprise swing in global economic activity (boom / bust)

Oil-specific demand shock

Precautionary / fear-driven shift in oil demand

Computing Betas, and Gammas

$$r^{(h)}_{s,t} = \alpha_{s,h} + \sum_k \beta_{s,k}(h) \cdot \epsilon^k_t + \sum_{l=1}^2 \theta_l \cdot \epsilon_{t-l} + \varphi \cdot r_{s,t-1} + \delta \cdot \text{controls}_t + u_{s,t+h}$$

with cumulative return $r^{(h)}_{s,t} = \sum_{j=0}^h r_{s,t+j}$

$$RV_{s,t+h} = \alpha_{s,h} + \sum_k \gamma_{s,k}(h) \cdot \epsilon^k_t + \sum_{l=1}^2 \theta_l \cdot \epsilon_{t-l} + \varphi \cdot RV_{s,t-1} + \delta \cdot \text{controls}_t + v_{s,t+h}$$

with realized vol $RV_{s,t} = \sqrt{(1/N_t) \sum_d r_{s,d}^2)} \times \sqrt{252}$

$k \in \{\text{supply, agg-demand, oil-specific}\}$; s = sector; h = horizon (0...12 months).

All four are estimated by OLS with HAC (Newey–West) standard errors; coefficients are kept only if $p < 0.10$.

β and γ are the objects of interest (return and volatility responses per 1σ shock).



Coefficients Robustness

1. HAC Standard Errors

Heteroskedasticity & Autocorrelation Consistent

Why?

Financial return residuals exhibit time-varying volatility and serial correlation. Plain OLS standard errors then under-state uncertainty, especially for cumulative LP regressions where Y observations overlap by construction.

Approach

Newey-West HAC covariance matrix with bandwidth $(h+1)$ at each horizon, matching the moving-average order induced by the cumulative residuals.

Benefit

Reliable t-statistics and confidence intervals; significance survives at cells with genuine systematic exposure.

2. MOPM Lag Augmentation

Montiel Olea & Plagborg-Møller (2021)

Why?

Persistent regressors (returns close to a random walk) can distort HAC inference. An alternative covariance with a different theoretical foundation provides a robust cross-check.

Approach

Add one extra lag to the LP regression and use HC1-robust (not HAC) standard errors. Theoretically valid under near-unit-roots and verified against the HAC baseline.

Benefit

Identifies HAC-fragile cells: when both methods agree, the conclusion is robust. ~12% of cells disagree at the borderline.

3. Wild Bootstrap CIs

Gonçalves-Kilian (2004); Pagan (1984) generated-regressor correction

Why?

The Stage-2 shocks are estimated, not observed. Naive HAC ignores that uncertainty, so reported significance is over-stated for the cells most exposed to first-stage noise.

Approach

500-iteration wild Rademacher bootstrap on the SVAR residuals. The full SVAR $\rightarrow$ LP pipeline re-runs on each perturbation; percentile CIs propagate first-stage noise into the betas.

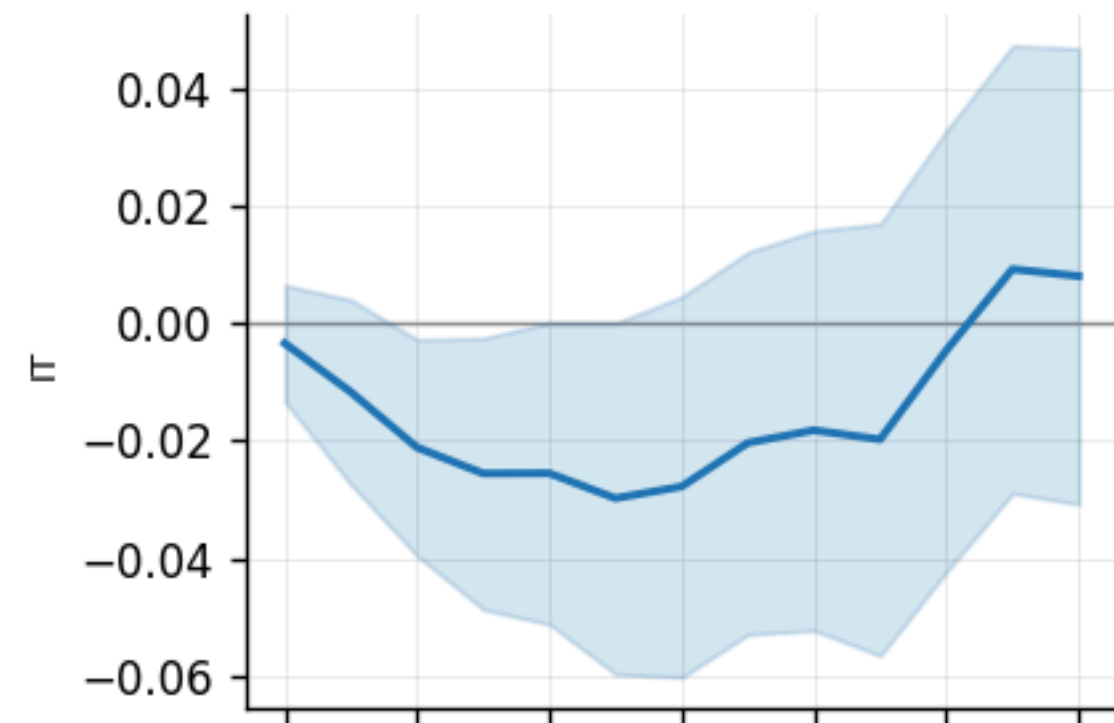
Benefit

Cells whose HAC significance does not survive the bootstrap are flagged and demoted. The surviving betas are robust to the generated-regressor problem.

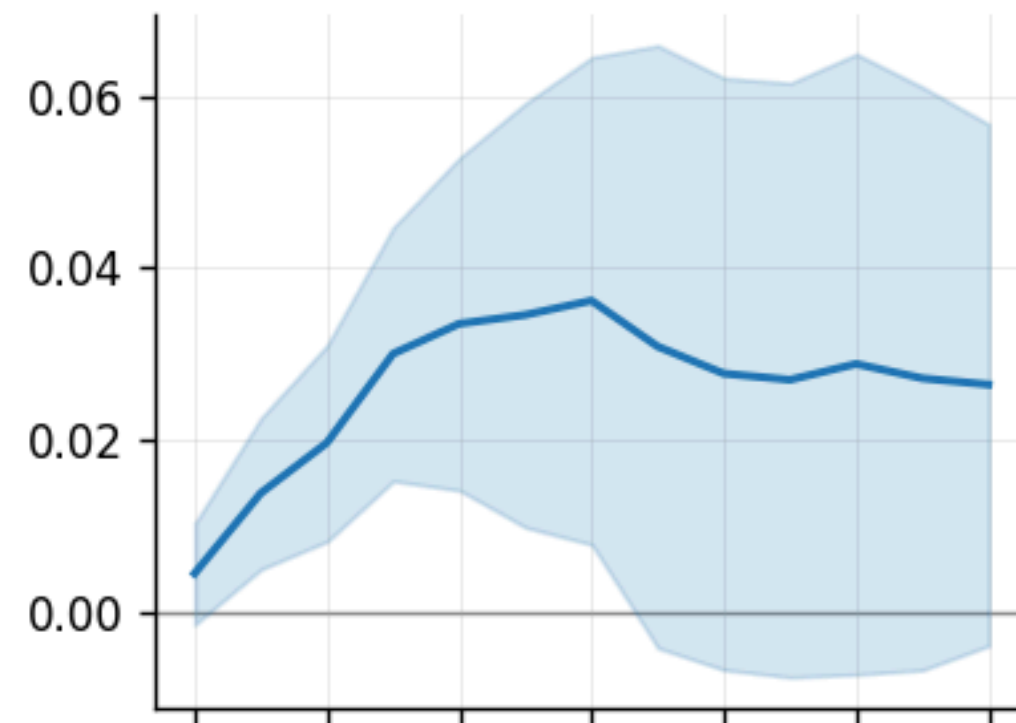
Risk management

Cumulative sectoral return response to Kilian structural shocks (90% HAC bands)

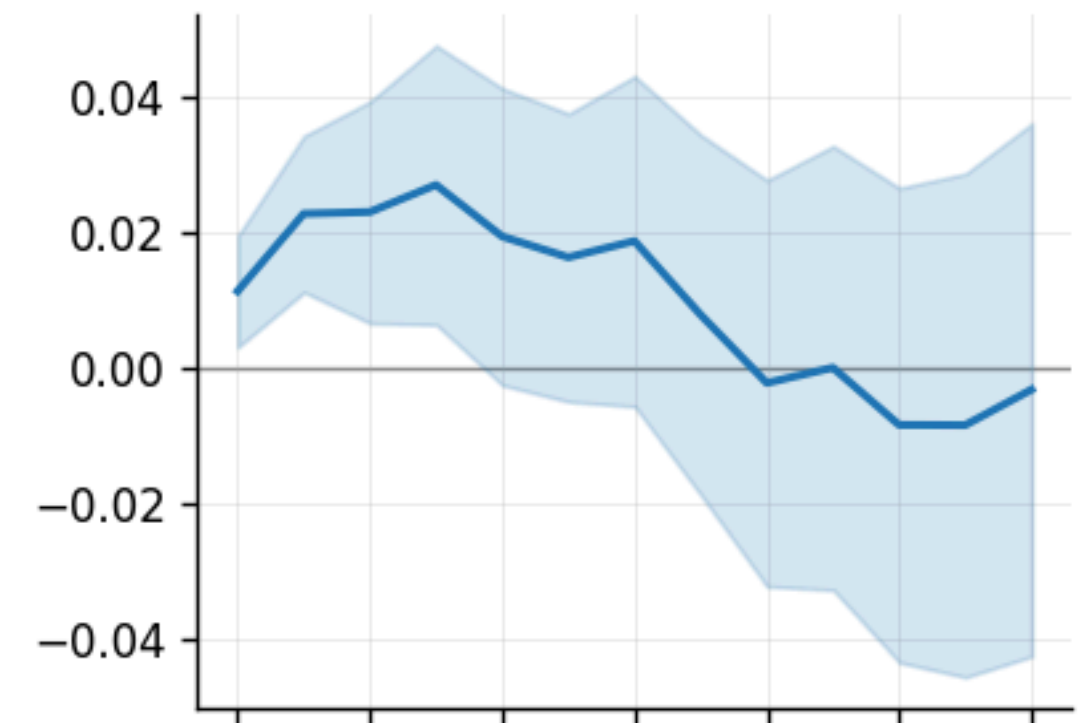
Supply Shock



Agg Demand Shock



Oil specific shock



Risk Management: The Baseline Portfolio

1. Why a baseline?

Oil-shock betas only tell us how to TILT — tilt relative to what?

- The signal is a RELATIVE bet across sectors
- We need a sensible starting allocation w^A before applying any oil-driven tilt
- Chosen baseline: risk parity (inverse-volatility weights)

2. Baseline: inverse-vol

Risk parity — less weight to volatile sectors

$$w_s^A = \frac{1/\sigma_s}{\sum_{s'} 1/\sigma_{s'}}$$

σ_s = historical sector volatility. Each sector contributes similar RISK, not similar capital.

3. Volatility targeting

Scale weights to a fixed annualised risk level

$$\sigma_{\text{port}}^2 = w' \Sigma w, \quad w^{\text{scaled}} = w \times \frac{\sigma_{\text{target}}}{\sqrt{w' \Sigma w}}$$

σ_{target} = 15% annualised. Variance is quadratic in w , so volatility scales linearly — calm markets → lever up; turbulent markets → delever.

4. Introducing the tilt

Linear adjustment — the simplest overlay

$$w_s^B = w_s^A + \lambda \cdot \frac{\text{signal}_s}{dv_s}$$

λ controls tilt intensity ($\lambda=0 \rightarrow$ base exactly). dv = downside volatility — risk-adjusts the tilt per sector.

The portfolio construction pipeline, one month at a time

Base weights w^A

Inverse-vol weights, vol-scaled to 15% target — computed BEFORE looking at any oil shock

+ Oil signal (this month)

z-scored, cross-sectional tilt direction from $\text{betas_mh} \times \tanh(\text{shocks})$

Tilted weights w^B

Linear or Black-Litterman tilt, renormalised to same gross exposure as w^A

Risk Management: The Black-Litterman Overlay

1. Prior: implied returns

“Reverse optimization” from the base portfolio

2. View & confidence

The oil signal becomes our “view”

3. Posterior blend

Bayesian precision-weighted average

4. Final weights

Back to mean-variance optimal weights

$$\Pi = \delta \Sigma w^A \quad (\delta = 2.5) \Rightarrow q = \text{signal} \times 0.02 \times \lambda, \quad \Omega^{-1} = \text{diag}\left(\frac{\text{betas_conf}}{\text{diag}(\Sigma)}\right) \Rightarrow \mu_{BL} = \left(\frac{1}{\tau}\Sigma^{-1} + \Omega^{-1}\right)^{-1} \left(\frac{1}{\tau}\Sigma^{-1}\Pi + \Omega^{-1}q\right) \Rightarrow w_{BL} = \frac{1}{\delta}\Sigma^{-1}\mu_{BL}, \quad w_{BL} \leftarrow \max(w_{BL}, 0)$$

“What returns would have to be expected for wA to be optimal?” $\delta=2.5$ is the standard BL risk-aversion constant.

betas_conf (total |t-stat| across horizons) sets Ω^{-1} — sectors with strong, reliable betas get high-confidence views.

Same logic as betas_mh (Step 1, earlier): blend prior Π and view q , weighted by their precisions. Weak views $\rightarrow$ stay near Π .

Long-only clip, then renormalised to wA’s gross exposure — this is wBL = wB from the previous slide.

Both overlays run side-by-side every month

B_lin

Simple, transparent — signal added directly via downside-vol scaling

B_bl

Theory-grounded — full covariance matrix + confidence-weighted views

$\lambda \rightarrow$ the signal’s reach

$\lambda=0 \rightarrow$ both reproduce wA exactly — the grid search over λ (next) picks the best tilt intensity

Risk management

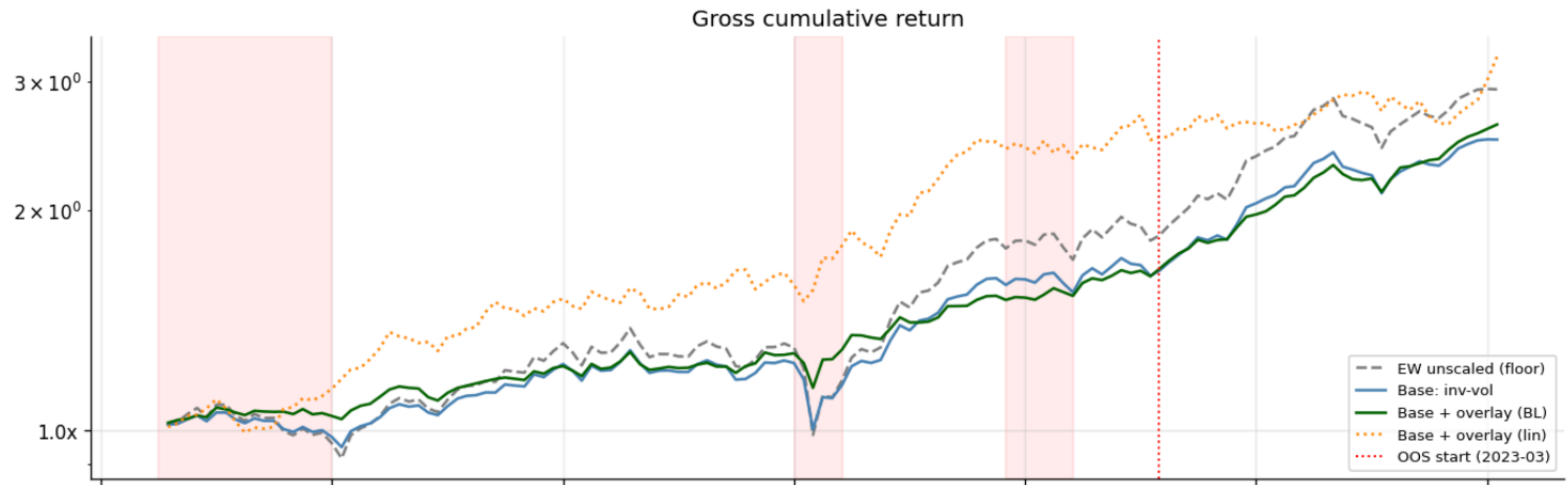
	Ann Ret %	Ann Vol %	Sharpe	Sortino	Max DD %
Strategy					
EW (unscaled floor)	16.43	9.54	1.721	2.279	-14.46
Base: inv-vol	14.69	8.00	1.836	2.337	-12.14
Base + oil overlay (BL)	16.14	6.16	2.618	3.033	-8.18

Performance Evaluation

		Oil-shock share %
Period	Strategy	
IS	Base: inv-vol	1.52
	Base + overlay (BL)	0.42
OOS	Base: inv-vol	1.50
	Base + overlay (BL)	0.40

Oil Shock Variance

Risk management



Risk management

Running capped backtest TILT_CAP=+-10% ...

Avg |active tilt|/sector: uncapped=0.0681 capped=0.0641
Oil-share (00S avg wt): base=1.5% uncapped=0.4% capped=0.4%
TE: uncapped=4.19% capped=3.95%

Episode ES(5%) – base $\angle$ uncapped $\angle$ capped:
2014–16: base=-2.56% uncapped=-1.64% (d=+0.91pp) capped=-1.74% (d=+0.81pp)
2020: base=-14.56% uncapped=-7.52% (d=+7.04pp) capped=-8.10% (d=+6.46pp)
2022: base=-3.17% uncapped=-1.27% (d=+1.89pp) capped=-1.43% (d=+1.74pp)

Verdict 2: PASS – capping preserves protection (3.0pp approx uncapped 3.3pp)

Fraction of months hedge is active: 65.5%

TE: always-on=4.19% conditional=3.56%

Oil-share (00S avg wt): base=1.5% conditional=0.9%

Episode ES(5%) – base $\angle$ always-on $\angle$ conditional:

2014–16: base=-2.56% always-on=-1.64% (d=+0.91pp) cond=-1.56% (d=+1.00pp) [12/18 months active]

2020: base=-14.56% always-on=-7.52% (d=+7.04pp) cond=-7.52% (d=+7.04pp) [1/6 months active]

2022: base=-3.17% always-on=-1.27% (d=+1.89pp) cond=-1.27% (d=+1.89pp) [8/8 months active]

THANK YOU !